

Boosting proton beam therapy potential

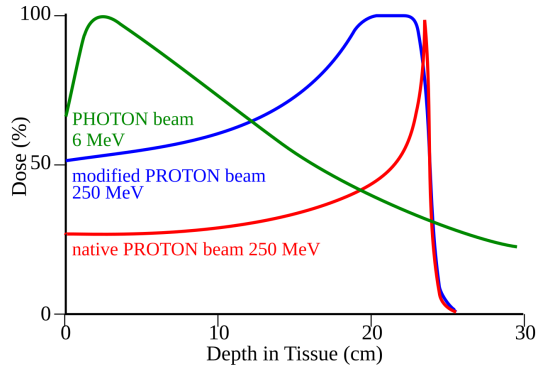
Clinical motivation and measurement principles

DIPC CRYSP group

Working draft — September 9, 2026

What is proton therapy?

- A form of **external-beam radiotherapy** that uses accelerated protons to destroy tumour cells.
- Protons are **positively charged particles**: their energy determines how deep they travel in tissue.
- Their dose peaks near the end of their range (the **Bragg peak**), then falls sharply.
- Combining different proton energies covers the tumour volume, while **reducing dose beyond it** compared with X-rays.



Schematic depth-dose profiles: a single proton energy, a spread-out peak and an X-ray beam.

Image: D. Ilyin / A. A. Miller, Wikimedia Commons (CC0).

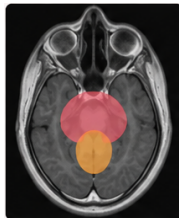
Reference: National Cancer Institute, External Beam Radiation Therapy.

Example: Brain tumours

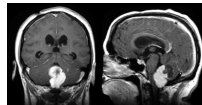
- Brain and CNS tumours are the **most common solid tumours in children**: about **300 new cases a year in Spain**.
- In adults, malignant brain/CNS tumours account for roughly **4,400 cases a year in Spain** (about 1.5% of cancers).
- Childhood ependymomas often arise near the **cerebellum and brainstem**. The goal is to cover the tumour while sparing these structures.

Childhood estimate: Pediatría Integral (2021), based on RETI.

Adult figure: approximate, from REDECAN 2026 all-age malignant CNS estimate.

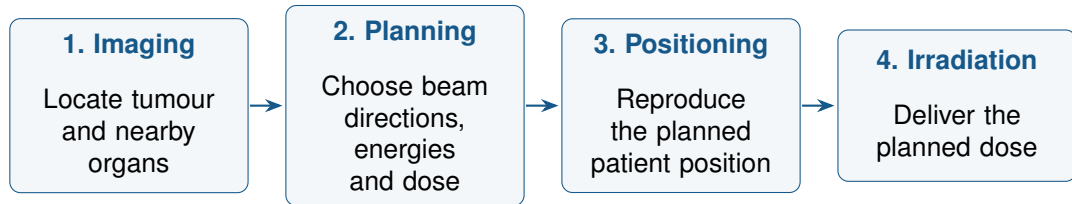


Brain/skull-base example: tumour (pink), brainstem (orange).



Fourth-ventricle ependymoma (MRI).

From treatment planning to delivery



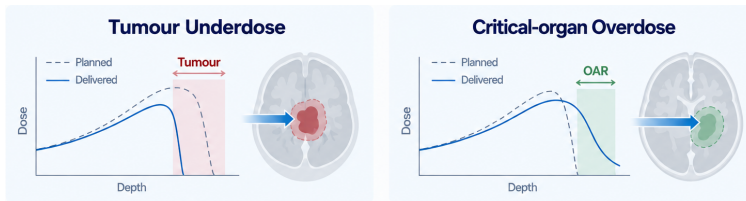
Delivered dose can differ from plan

- Uncertainty in tissue stopping power.
- Anatomical changes, positioning and motion.

Unlike photons, protons are absorbed in the patient's body

The plan delivers a **prediction** of proton range. **How to check it?**

Why range uncertainty matters



- Systematic errors in proton range **estimation** are of the order of **3.5%**.
- The required planning margins are of the order of several mm.
- Conservative beam choices protect nearby organs, but can limit coverage when tumour and organ are close.

Example at 10 cm depth

$$\begin{aligned} 0.035 \times 100 \text{ mm} + 1 \text{ mm} \\ = 4.5 \text{ mm} \end{aligned}$$

3.5% range allowance plus 1 mm.
Margins depend on the centre and treatment site.

Range uncertainty: Paganetti, Phys. Med. Biol. 57, R99 (2012). OAR = organ at risk.

What could better range verification change?

- With less range uncertainty, planners can reduce dose to **healthy tissue near the tumour**.
- They may also use **beam directions otherwise avoided** because a critical organ lies near the stopping point.
- Verification must show that the uncertainty is sufficiently small before these changes can be adopted.

A planning study: 27 brain-tumour patients

Range uncertainty reduced from **3% to 2%**.

24 of 27 patients (89%) had a relevant dose reduction to at least one organ at risk.

In brain tumours, proton range uncertainties of **1 mm or less** can in principle be achieved.*

Taasti et al., British Journal of Radiology 96, 20230110 (2023).

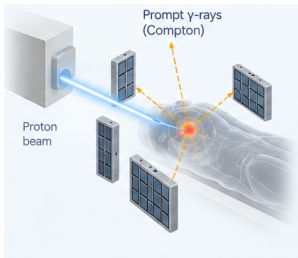
Beam-direction benefit: Tattenberg et al., Medical Physics (2022).

* J.J. Gómez-Cadenas et al., *A High-Sensitivity In-Room CRYSP Scanner for Distal-Edge Verification in Proton Therapy*, paper in preparation, available upon request.

Complementary imaging for proton therapy

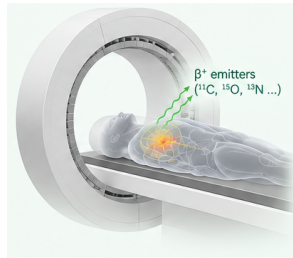
Compton camera

In-beam: prompt γ rays



PET

In-room, post-treatment: β^+ activity



R&D studies support Compton cameras and PET for **proton range verification**. No dedicated commercial solution identified.

Safavi et al., Nat. Commun. 17, 206 (2026).

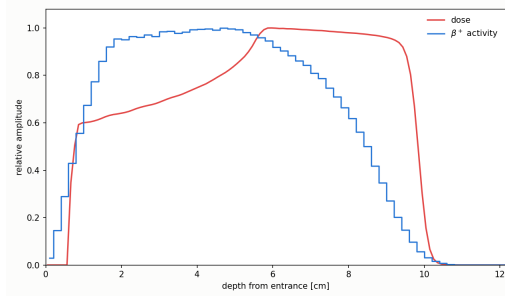
Sharma et al., Med. Phys. 52, 1123 (2025).

Parodi et al., Eur. J. Nucl. Med. Mol. Imaging 51, 183 (2024).

Zhu et al., Phys. Med. Biol. 56, 6425 (2011).

PET: an *in vivo* probe of where the beam stopped

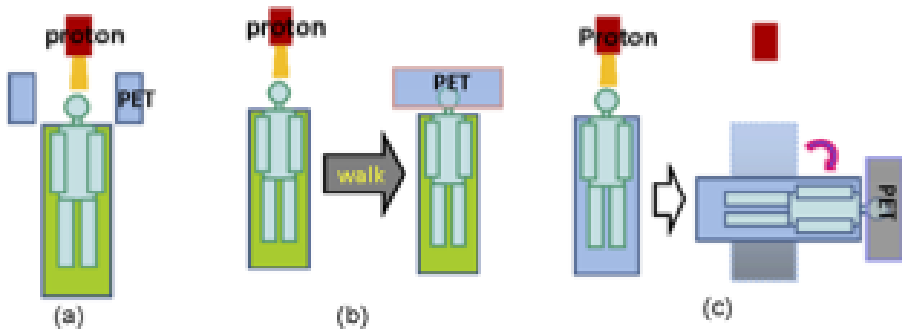
- Above the nuclear-reaction threshold, protons create β^+ emitters along their path — mainly ^{15}O and ^{11}C .
- Their annihilation γ 's are imaged in coincidence: a PET image of the induced activity.
- The activity falloff sits just upstream of the dose edge; its distal edge tracks the range.



Illustrative dose and β^+ activity profiles.

The activity edge is compared with a prediction to estimate the range.

PET range verification: offline, in-beam and in-room



PET range verification: offline, in-beam and in-room

Offline

patient moved to
a nearby PET/CT

15–30 min later

washout, ^{15}O gone,
co-registration

In-beam

two planar heads
at the treatment
position

~ zero delay

limited angular
coverage

In-room

full ring, next
to the treatment
room

~ 2 min delay

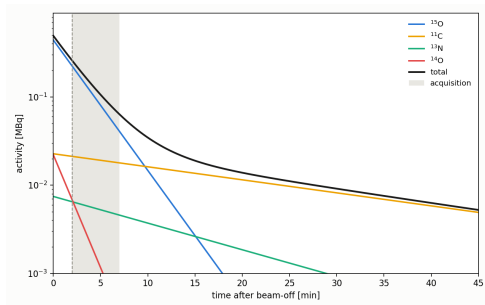
full coverage,
high sensitivity

The trade-off is **delay** vs. **angular coverage** / **scanner quality**.

Offline results in large uncertainties

A delay of 15–30 minutes costs on three fronts:

- **Sensitivity**: the abundant ^{15}O (2 min) has vanished — only the sparse ^{11}C tail remains.
- **Biological washout**: perfusion clears the emitters; hard to model, patient-dependent.
- **Co-registration**: the patient is set up afresh on a separate scanner — a few-mm misalignment maps straight into an apparent range shift.



^{15}O dominates the first minutes, then decays; ^{11}C carries the late, weak signal.

Online/In-room tradeoffs

In-beam (short delay, same bed)

- Retains the short-lived ^{15}O , essentially no delay.
- But the beam must pass: only **two planar heads** fit.
- Reduced angular coverage $\Rightarrow$ **low sensitivity** and limited-angle image artefacts.

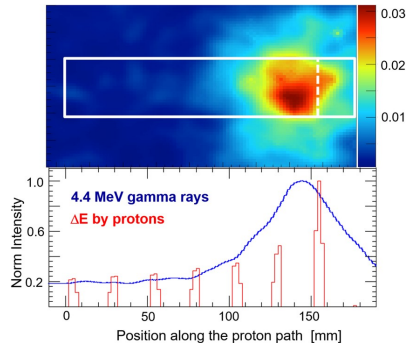
In-room (just after, full ring)

- A **full closed ring** $\Rightarrow$ full coverage, high sensitivity, clean image.
- Same couch $\Rightarrow$ a known rigid transform and a short transfer, without separate-scanner repositioning.
- The ~ 2 -min delay is **bought back** by sensitivity and a **better scanner**.

A **high-sensitivity in-room ring** combines short transfer time with complete angular coverage and high sensitivity.

Prompt gammas: tracking the proton range during irradiation

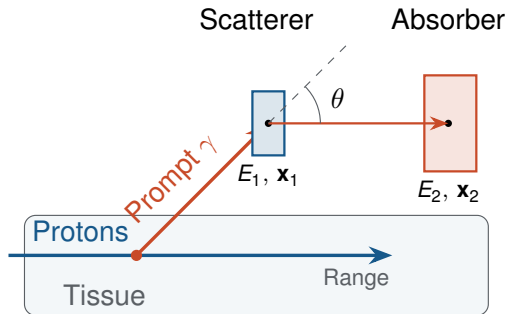
- Nuclear reactions along the proton path leave excited nuclei in the tissue. Their de-excitation emits **prompt γ rays** with energies of several MeV.
- Emission occurs **during irradiation**. Its distal falloff tracks the proton range; the relation depends on tissue composition and the selected γ energies.
- The measured **4.4 MeV γ profile** shown here concentrates near the Bragg peak and follows the proton range.



Measured prompt- γ profile and simulated proton energy deposition. 70 MeV protons; spaced PMMA slabs.

Imaging prompt gammas with a Compton camera

- A **Compton camera** measures the positions and energies of successive interactions of the **same γ ray**.
- The photon scatters in the first detector and is absorbed in the second. The deposited energies determine the **Compton scattering angle**.
- The two interaction positions define the cone axis; the scattering angle defines its opening. The emission point lies on this **cone of possible origins**.
- Combining many events reconstructs the **emission profile and its distal edge**.



One photon, two detector interactions: Compton scattering followed by absorption.

Hueso-González et al., Front. Oncol. 6:80 (2016).

Hybrid in-beam PET and Compton imaging (2022).

PET and Compton imaging: complementary measurements

	Prompt gammas / Compton	Induced activity / PET
Signal	Single prompt γ rays, several MeV	Pairs of 511 keV annihilation photons
Timing	During beam delivery	In the proposed in-room scheme, shortly after irradiation
Information	Emission profile along the beam; comparison with expected range	Activity distribution; comparison with expected distal edge
Main limitation	Few useful multi-interaction events and beam-related background	Decay, biological washout and patient registration

DIPC proposed strategy: Compton imaging for monitoring during irradiation, followed by an in-room PET measurement within minutes of beam stop. Combine both results to minimize systematic errors.

CRYSP: PET with cryogenic cesium iodide

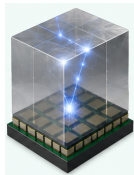
Pure Csl at ~ 100 K



$\sim 10^5$ photons/MeV.

Abundant material; lower crystal cost than rare-earth scintillators.

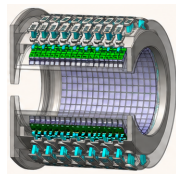
Monolithic 3D detection



Large crystals read out by silicon photomultipliers (SiPMs).

Millimetre-scale 3D resolution in simulation; measured energy resolution below 7% at 511 keV.

Full-ring PET

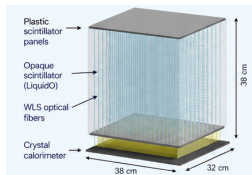


Total-body PET design based on these crystals.

High sensitivity at lower detector cost is the design goal.

COCOA: a compact Compton camera for MeV gamma rays

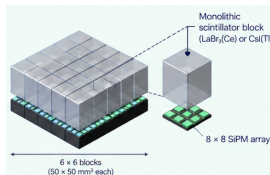
LiquidO scatterer



Opaque liquid scintillator with a 3D grid of wavelength-shifting fibres.

Reconstructs the **position and energy** of the first interaction.

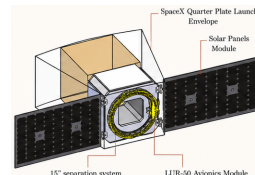
Crystal absorber



Monolithic scintillator blocks read out by SiPM arrays.

Measures the scattered photon: **3D position and energy**, including depth of interaction.

Space application

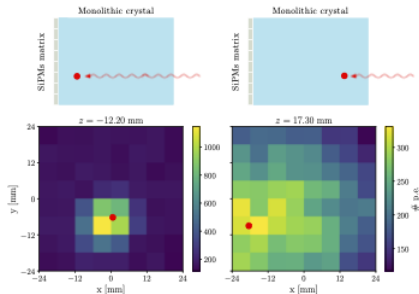


Designed for **MeV gamma-ray astronomy**.

Compact geometry for microsatellites or high-altitude balloons; proposed LUR-50 integration shown.

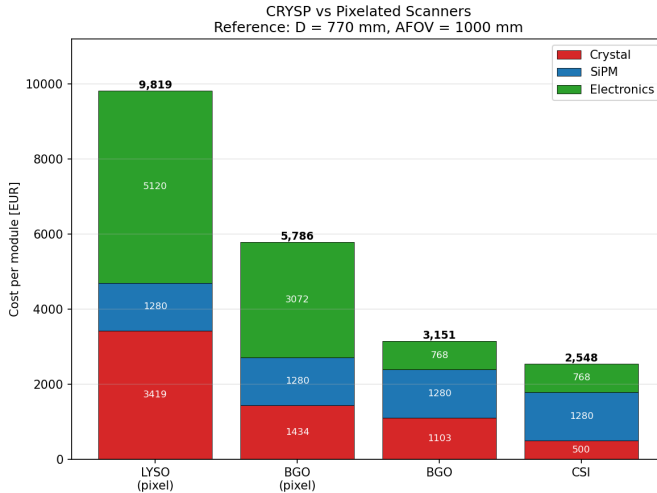
CRYSP: *CRYogenic Sensitive PET*

- Scintillators **CsI** and **BGO** gain dramatically in light yield when cooled — at 77 K, CsI reaches ~ 100 photons/keV and BGO ~ 29 photons/keV.
- In both cases, high yields translate into excellent energy resolution (6% CsI, 10% BGO), which in turn is essential for efficient scatter rejection.
- Monolithic crystals + SiPMs permit 3D position reconstruction using neural networks with high accuracy and no parallax error.



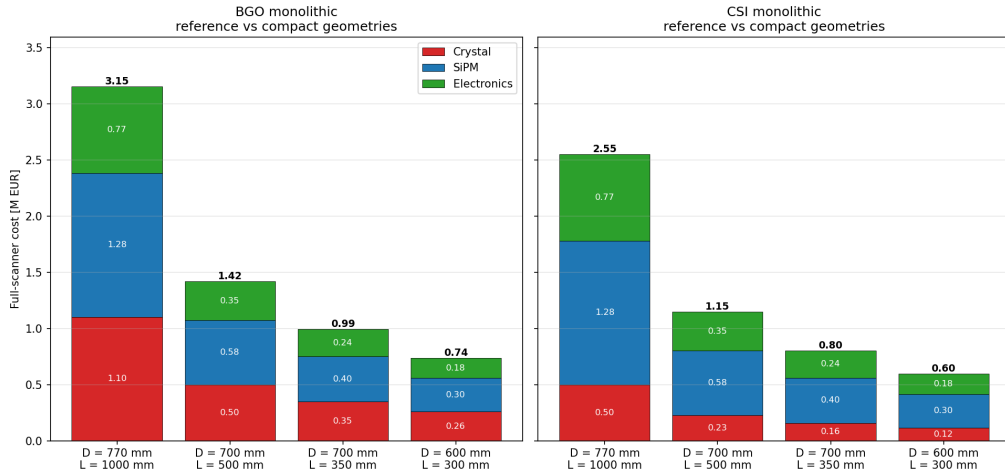
3D reconstruction in a monolithic crystal.

Scanner cost breakdown



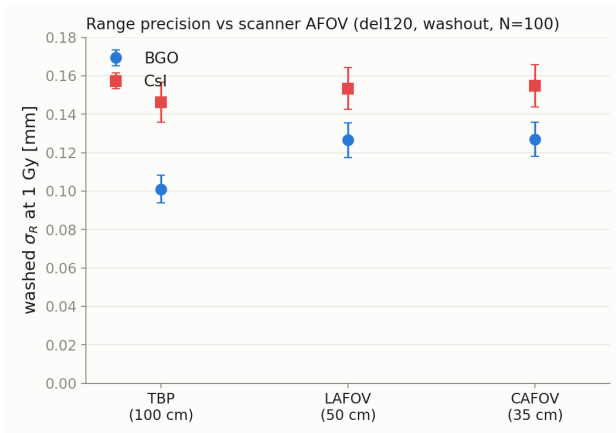
Total scanner cost

CRYSP Scanner cost breakdown



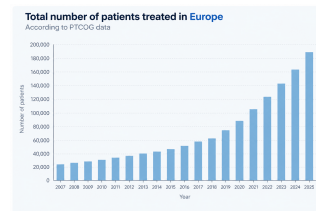
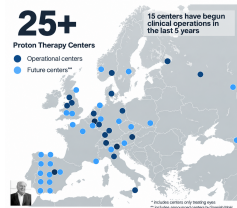
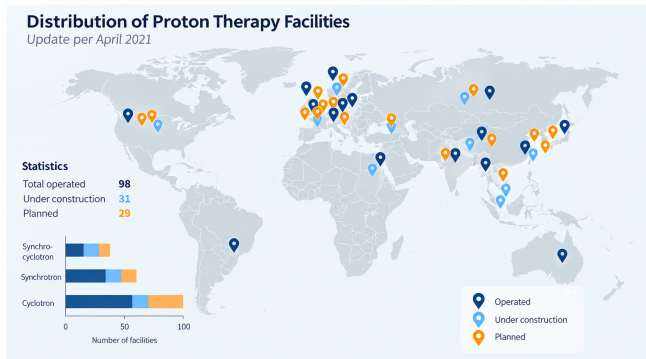
Reference protocol: all scanner configurations are sub-mm

- **Protocol:** 1 Gy, 300 s acquisition, starting 120 s after beam-off; washout included.
- BGO: 0.101–0.127 mm; Csl: 0.146–0.155 mm.
- BGO is more precise through higher sensitivity; Csl remains a competitive, lower-cost option.
- These values are the **statistical precision of the fitted activity edge**.



Finite-pool-corrected σ_R for the reference protocol ($N = 100$).

Proton therapy: worldwide deployment and growth in Europe

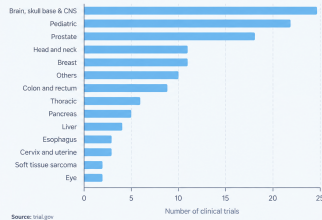


Over the last two decades, proton therapy has evolved from a highly specialised treatment into a **globally deployed clinical technology**.

World map: April 2021. European patient chart: PTCOG attribution in the supplied figure; numerical series not independently verified.

Clinical evidence for proton therapy

Increasing clinical evidence for indication



Majority in CNS, pediatric, prostate, breast, and lung

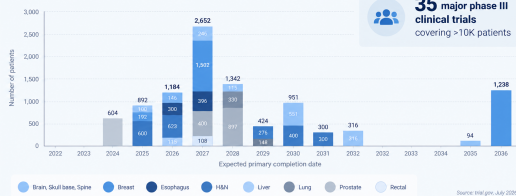


Mainly focused on side-effects reduction

Expanding clinical evidence pipeline

Supporting momentum for proton therapy

Number of patients per ongoing phase III clinical trial (proton therapy vs. radiotherapy)



35 major phase III clinical trials covering >10K patients

Phase III trials are building the **clinical evidence** for wider use of proton therapy, particularly to **reduce side effects**.

Supplied charts cite trial.gov (phase III chart: July 2026). Trial counts and projected completion dates require registry verification.

Boosting proton therapy

1. Proton beam dose monitoring **in real time** is a real need that will become more pressing as the number of centres grows and the number of treated patients increases.
2. R&D on monitoring techniques has been going on for about 20 years. PET and gamma cameras offer the potential for **day-by-day, case-by-case monitoring** and distal-edge measurements that could reduce underdose while minimising the risk of overdose.
3. However, these systems have **not yet become standard medical practice**.

Clinical status: Current status and upcoming developments for online adaptive proton therapy, Front. Oncol. (2025).

The DIPC group

The DIPC group is a leading neutrino physics group in Spain, with extensive expertise in **nuclear instrumentation**.

Scientific leadership and concepts

- Led by **J.J. Gomez-Cadenas**: co-recipient of the **2016 Breakthrough Prize in Fundamental Physics** as a member of K2K and T2K, with group leadership roles in neutrino physics.
- Two ERC grants: **AdG-2013** and **SyG-2020**.
- **CRYSP**: Gomez-Cadenas, Soleti, Lopez-Guejo, Pelegrin.
COCOA: Soleti, Gomez-Cadenas.

Team, engineering and management

- Four faculty, two engineers, two postdocs, and graduate and undergraduate students.
- Substantial experience in mechanical engineering and cryogenics.
- Strong Basque industry connections and established management: Lopez-Guejo and Pelegrin.
- Well funded by the ERC and Basque Government; additional positions planned for CRYSP and COCOA.

CRYSP and COCOA for proton beam monitoring

- The CRYSP and COCOA concepts can be applied as **high-performance, low-cost solutions** for proton beam monitoring. Our recent study shows the potential of in-room CRYSP PET.
- We have **operating prototypes of both systems** and are moving towards the construction of larger apparatus.
- The group is establishing a close collaboration with **Biogipuzkoa** and the medical staff of the forthcoming proton beam therapy centre in **Donostia**.

J.J. Gomez-Cadenas et al., *A High-Sensitivity In-Room CRYSP Scanner for Distal-Edge Verification in Proton Therapy*, paper in preparation, available upon request.

Our vision: an integrated monitoring service

- Optimised, **high-precision, low-cost apparatus**, combined with fast and efficient software using **AI-enabled technologies**, to provide real-time monitoring and support treatment correction in cooperation with doctors.
- A potential spin-off, **ARGOS**, would offer an integrated solution to proton therapy centres:

Hardware

Installation and maintenance

Software

Full development and
integration

Day-by-day service

Close cooperation with
doctors

The model could be extensively tested and optimised in **Donostia**, then expanded to new proton beam centres.

Development programme and investment

Foreseen development cycle: 3–4 years.

Investment: approximately €1 million per year, or €3–4 million in total.

1. Validation of prototypes.
2. Preclinical demonstrators.
3. Clinical trials and medical validation.
4. Construction of medical-grade apparatus.
5. Initial operation.